

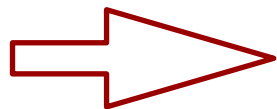
Peter Willendrup

Establishing the learning goals, a look at the programme

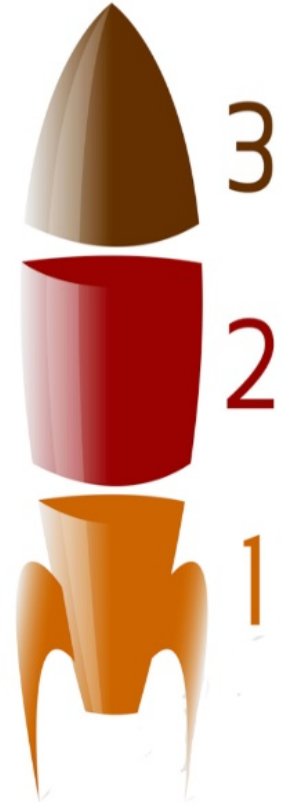
Learning goals:



1. Learn McStas basics
2. Build and operate simple instrument models, source + optics + sample + detector
3. Learn basics of instrument-optimisation and resolution calculations
5. Add Mantid / NeXus capabilities
6. Get a better idea of what you want to do with McStas, how to do it, how to get help
7. Get up-to-speed with latest developments and advanced features of McStas and related tools like NCrystal and McStasScript.



Enable your independent work with McStas





McStas

November 4th Beginners McStas	Time (GMT)	November 5th Instrument design	Time (GMT)	November 6th Advanced A	November 6th Advanced B
Welcome + setting learning goals 30 min McStas intro + general concepts 30 min McStas live demo / "Build-along"	9:00-10:30	30 min Polarisation 60 min Lesser known tips and tricks, including: - Tools for optimisation	9:00-10:30	Advanced topics: NCrystal talk Using NCrystal in McStas & McStasScript	Work on your own instrument w / help and guidance
Responsible: Peter		Responsibles: Peter		Responsible: Thomas	
Coffee Break	10:30-11:00	Coffee Break	10:30-11:00		Break
60min Guides and gravity: 20 min presentation 40 min practical	11:00-12:30	90 min McStasScript intro/exercise	11:00-12:30	Advanced topics interactive: Write your first component	Work on your own instrument w / help and guidance
Responsible: Mads		Session lead: Mads		Responsible: Peter	
Lunch break	12:30-13:30	Lunch break	12:30-13:30	Lunch break	Lunch break
Choppers / rotating optics: 20 min presentation 25 min practical 45 min Sample + Union overview talk Responsibles: Mads + Peter	13:30-15:00	McStas -> Mantid, NeXus: 20 min presentation 20 min demo 50 min exercise Responsible: Peter	13:30-15:00	Advanced talk/demo: McStas Union for detectors	Work on your own instrument w / help and guidance
Coffee Break	15:00-15:30	Coffee Break	15:00-15:30	Coffee Break	Break
45 min Sample exercise 45 min Union exercise Responsibles: Peter + Mads	15:30-17:00	Time of flight resolution, talk + exercise Responsible: Mads	15:30-17:00	Porting old comps and instruments to McStas 3 (and allow GPU execution)	Work on your own instrument w / help and guidance



Links to various infrastructure and services at:
<https://isis2025.mcstas.org>

School programme - day 1



November 4th Beginners McStas

Welcome + setting learning goals
30 min McStas intro + general concepts
30 min McStas live demo / "Build-along"

Responsible: Peter

Coffee Break

60min Guides and gravity:
20 min presentation
40 min practical

Responsible: Mads

Lunch break

Choppers / rotating optics:
20 min presentation
25 min practical
45 min Sample + Union overview talk
Responsibles: Mads + Peter

Coffee Break

45 min Sample exercise
45 min Union exercise

Responsibles: Peter + Mads

Intro lecture, general principles,
sources + monitors,

Lectures + "recipe" exercises

Sample-lecture, including "advanced McStas
grammar", sample exercises

In "cookbook" sections,
think ahead toward
your own project:

- * Which neutron source
- * What optics
- * What sample

- K.I.S.S. for now

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School programme - day 2



November 5th Instrument design

30 min Polarisation
60 min Lesser known tips and tricks, including:
- Tools for optimisation

Lectures on polarisation
and instrument optimisation
technicals

Responsibles: Peter

Coffee Break

90 min McStasScript intro/exercise

ToF resolution, talk + exercise

Session lead: Mads

Lunch break

McStas -> Mantid, NeXus:

20 min presentation

20 min demo

50 min exercise

Responsible: Peter

Mantid-howto,
lecture and demo

Coffee Break

Time of flight resolution, talk + exercise

McStasScript intro/exercise

Responsible: Mads

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School programme - day 3, advanced stuff + work on own project



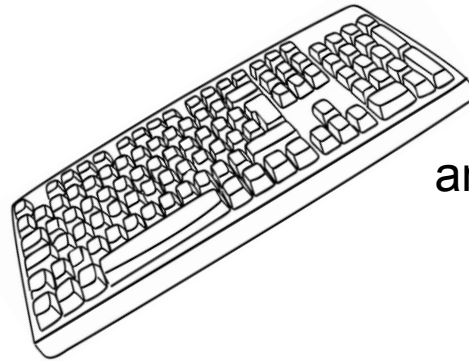
November 6th Advanced A	November 6th Advanced B
Advanced topics: NCrystal talk Using NCrystal in McStas & McStasscript Responsible: Thomas	Work on your own instrument w / help and guidance
	Break
Advanced topics interactive: Write your first component Responsible: Peter	Work on your own instrument w / help and guidance
Lunch break	Lunch break
Advanced talk/demo: McStas Union for detectors Responsible: Mads	Work on your own instrument w / help and guidance
Coffee Break	Break
Porting old comps and instruments to McStas 3 (and allow GPU execution) Responsible: Peter	Work on your own instrument w / help and guidance



For the exercise-based work-sessions

- You will benefit from working in pairs, $2 > 1$

- Take turns being the “coder”



and the “parallel processor”



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Let's get to it!



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